

RESEARCH ARTICLE

Enhancing Spatial-Temporal Crime Forecasting Using Machine Learning for Law Enforcement

NANAN SOEKARNA¹, TEDDY MANTORO¹, (Senior Member, IEEE), UMAR ADITIAWARMAN¹,
FIRMAN ARIFMAN¹, (Member, IEEE), AND BEN RAHMAN²

¹School of Computer Science, Nusa Putra University, Jakarta, Indonesia

²Department of Information Systems, Universitas Nasional, Jakarta, Indonesia

Corresponding author: Firman Arifman (firman.arifman@nusaputra.ac.id)

ABSTRACT Traditional crime forecasting often lacks the granularity and adaptability required to address evolving spatial and temporal crime patterns. Standard reactive models may fail to capture the complex, non-linear dynamics of urban crime, leaving law enforcement with limited insights for proactive resource allocation. This study does not aim to introduce a new machine learning algorithm; rather, it develops and evaluates an applied district-level spatio-temporal crime forecasting workflow for Bekasi City, Indonesia. The proposed workflow integrates Density-Based Spatial Clustering of Applications with Noise (DBSCAN) for spatial hotspot identification and Long Short-Term Memory (LSTM) neural networks for temporal sequence forecasting. Utilizing a five-year dataset (2020–2024) of street crime in Bekasi City, the methodology incorporates district-level demographic data from Statistics Indonesia and geocoded incident coordinates to model crime at a granular administrative scale. The framework identifies persistent and emerging hotspots, forecasts daily crime counts, and explores crime-category classification as a secondary analytical task. Results show that forecasting performance varies across districts, with stronger performance in relatively stable districts such as Pondokmelati (MAE: 0.24, MAPE: 10.1%) and weaker performance in sparse or volatile districts. The demographic analysis suggests a positive association between population density and cumulative crime incidence, although the finding should be interpreted cautiously because it is based on a limited number of districts. Rather than claiming causal validation of criminological theory, the results are interpreted as being consistent with Crime Pattern Theory and Routine Activity Theory in the observed Bekasi context. This study contributes a context-specific, data-driven decision-support workflow that combines spatial clustering, temporal forecasting, and forecast-informed visualization for aggregate law-enforcement planning. By translating model outputs into visual analytical summaries, the framework provides supporting evidence for district-level patrol planning and urban public-safety analysis, while emphasizing the need for human-in-the-loop interpretation and ethical safeguards in any operational deployment.

INDEX TERMS Crime forecasting, spatial-temporal analysis, machine learning, DBSCAN, LSTM, hotspot detection.

I. INTRODUCTION

Crime forecasting plays a critical role in supporting proactive policing and enabling data-driven allocation of law enforcement resources. In Indonesia, while national crime trends have shown some improvement, urban centres such as

Bekasi City continue to experience localized fluctuations in crime types and frequency. These fluctuations are influenced by a range of socio-economic, demographic, and environmental factors, including urban density, migration patterns, income disparity, and the distribution of law enforcement infrastructure.

Indonesia's urbanization rate has increased significantly in recent years, contributing to population concentration in

The associate editor coordinating the review of this manuscript and approving it for publication was Ayman El-Baz¹.

metropolitan and peri-urban regions, such as Bekasi. With over 2.5 million residents and its proximity to Jakarta, Bekasi has rapidly transformed into a densely populated commuter hub. This transformation has not only increased economic activity but has also contributed to shifts in crime patterns, particularly in street-level offenses, such as theft, fraud, and assault. Traditional statistical models that rely on historical averages or linear assumptions often fail to accommodate the nonlinear, multidimensional nature of such crimes.

In response to these challenges, predictive policing has emerged as a promising strategy that leverages data-driven tools to anticipate crime occurrence before it happens. However, many existing models fall short in two critical dimensions: spatial precision and temporal resolution. They often aggregate data at the city or regional level, thereby diluting micro-level crime patterns and operating with limited adaptability to changing trends. Consequently, law enforcement agencies lack actionable insights that can inform tactical operations on a daily or weekly basis.

Although spatial clustering, temporal forecasting, and hybrid spatio-temporal crime prediction models have been widely studied, their operational integration at the district level remains an important applied challenge, particularly in local urban contexts where law-enforcement decisions are made using administrative boundaries. Existing studies have demonstrated the usefulness of methods such as DBSCAN and LSTM; therefore, this study does not claim to introduce a new machine learning algorithm. Instead, it focuses on developing and evaluating an applied district-level spatio-temporal crime forecasting workflow for Bekasi City, Indonesia.

This study addresses these gaps by integrating established machine learning techniques: spatial clustering using DBSCAN and temporal prediction using LSTM networks. DBSCAN allows the detection of crime hotspots that may be irregular in shape and dynamic over time, whereas LSTM offers the capability to model sequential dependencies in crime occurrences. The integration of these techniques is intended to support district-level operational interpretation by linking the “where” and “when” dimensions of crime occurrence, rather than to propose a fundamentally new spatio-temporal learning architecture.

Furthermore, the proposed workflow is interpreted in relation to criminological theory. Crime Pattern Theory and Routine Activity Theory both emphasize the structured nature of crime, highlighting the importance of environmental design and temporal routines in influencing criminal behaviour. Rather than claiming to validate these theories causally, this study examines whether the observed spatial and temporal patterns are consistent with these theoretical perspectives in the Bekasi context. By aligning machine learning outputs with theoretical and practical considerations, this study aims to support law enforcement agencies in aggregate patrol planning, hotspot interpretation, and public-safety analysis, while recognizing the need for human judgment and ethical safeguards in any operational deployment.

The main contribution of this study is therefore applied and contextual rather than purely algorithmic. Specifically, this study contributes: (1) a district-level spatio-temporal crime forecasting workflow that integrates DBSCAN-based hotspot detection and LSTM-based daily crime-count forecasting using real-world police data from Bekasi City; (2) an empirical evaluation of forecasting performance across districts with different crime-pattern stability; (3) exploratory crime-category classification and forecast-informed hotspot visualization as supporting components for operational interpretation; and (4) a responsible decision-support framing that positions the model outputs as aggregate analytical signals rather than automated enforcement decisions.

In summary, the objective of this study is to design, implement, and evaluate an applied district-level spatio-temporal crime forecasting workflow for Bekasi City. Through granular analysis, comparative evaluation, and visual interpretation, the workflow aims to provide context-specific decision-support evidence for crime analysis in complex urban environments. The remainder of this paper is organized as follows: Section II reviews the relevant literature; Section III details the methodology, including data preparation, feature engineering, and model development; Section IV presents the results; and Section V discusses the conclusions, implications, and directions for future research.

II. RELATED WORK

Over the past two decades, crime forecasting has evolved from simple statistical projections to sophisticated, data-driven systems that incorporate artificial intelligence, geographic information systems (GIS), and real-time data streams [1], [2], [3], [4]. The primary objective of these systems is to support law enforcement with accurate predictions of when and where crimes are likely to occur, thereby enabling proactive intervention and resource deployment [5], [6]. Despite significant advancements, several challenges persist, particularly in integrating spatial and temporal analyses at granular geographic levels. In particular, the challenge is not only to improve predictive accuracy, but also to design workflows that remain interpretable, reproducible, and usable for district-level operational planning.

A. TEMPORAL FORECASTING MODELS

Classical time-series models, such as the Autoregressive Integrated Moving Average (ARIMA), have traditionally been used to model crime trends [7], [8], [9], [10]. Although effective in modelling linear temporal dependencies, these models are limited in their ability to capture the nonlinear and long-range dependencies often observed in real-world crime data. To overcome these limitations, researchers have turned to Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks. LSTM models have shown considerable promise in modelling sequential data owing to their capability to learn from long-term dependencies and manage vanishing gradient problems [11], [12], [13], [14].

Wang et al. [15] demonstrated that LSTM models outperform traditional forecasting approaches for urban crime prediction in terms of MAE and RMSE. Similarly, Kang and Kang [16] utilized multimodal LSTM frameworks to forecast different categories of crime incidents and reported improved predictive performance across several datasets. These studies underscore the suitability of LSTM for crime forecasting tasks, particularly when fine temporal resolution and model adaptability are critical. However, because LSTM is already an established temporal forecasting method, the present study does not position LSTM itself as a methodological novelty. Instead, LSTM is used as one component within an applied district-level forecasting workflow.

B. SPATIAL CLUSTERING AND HOTSPOT ANALYSIS

Hotspot analysis has long been a cornerstone of crime-prevention strategies. Common techniques include Kernel Density Estimation (KDE), K-Means clustering [17], and DBSCAN. Among these, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), introduced by Ester et al. [18], is particularly effective for crime data because of its ability to detect arbitrarily shaped clusters and filter noise. DBSCAN does not require the number of clusters to be predefined, making it well-suited for dynamic environments such as urban crime settings.

Yu et al. [19] applied DBSCAN to crime data in metropolitan areas and demonstrated its competitive performance in identifying hotspots compared to traditional methods. Leitner and Helbich [20] and Chainey and Ratcliffe [21] also highlighted the operational benefits of DBSCAN for urban policing. However, in most of these studies, spatial clustering was performed independently of forecasting, thus missing the opportunity to integrate spatial dynamics into predictive models. As with LSTM, DBSCAN is not treated in this study as a new algorithmic contribution. Its role is to provide an interpretable spatial representation of hotspot structures that can be connected with temporal forecasting outputs for district-level analysis.

C. INTEGRATED SPATIO-TEMPORAL MODELS

Several recent studies have attempted to combine spatial and temporal components to improve the accuracy of crime forecasting. Zhao et al. [22] integrated DBSCAN-based clustering with ARIMA models to capture spatial structures and temporal trends simultaneously. Their findings indicate significant improvements in predictive performance. In a more advanced approach, Shan et al. [23] proposed Ada-GCNLSTM, a hybrid model combining Graph Convolutional Networks (GCNs) with LSTM layers, resulting in a reduced MAE by 11.7% across multiple urban datasets.

Despite these innovations, integrated models often struggle to maintain the granularity and interpretability required for tactical law enforcement. As the spatial and temporal granularity becomes finer, data sparsity increases and model interpretability decreases [24], [25]. Machine learning-based

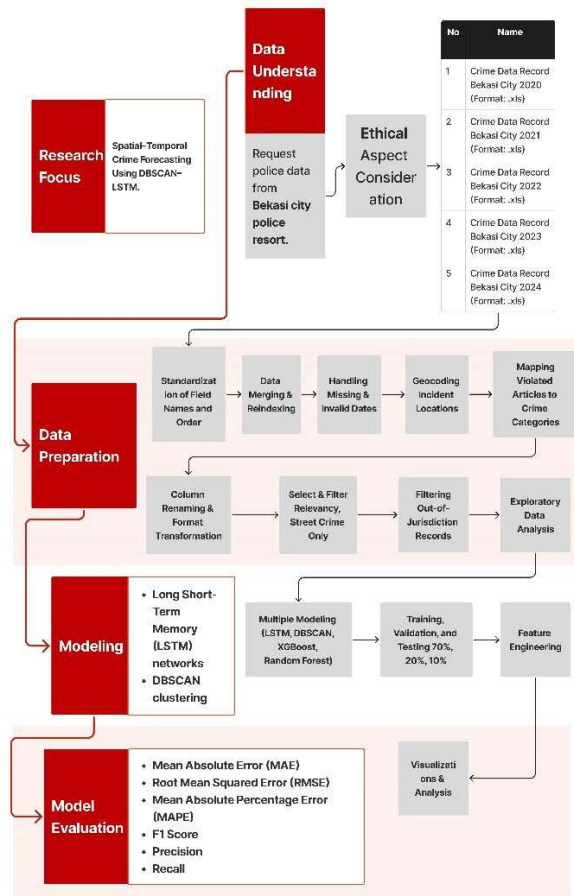


FIGURE 1. CRISP-DM research framework.

prediction models often suffer from a ‘black-box’ nature, where improved performance comes at the cost of interpretability [24]. Furthermore, most implementations focus on overall crime counts at macro scales, neglecting the importance of crime category prediction, which is vital for targeted intervention [26]. District-level crime prediction models that support tactical law enforcement deployment remain underdeveloped compared to macro-level policy-focused models [27], [28]. Therefore, the relevant research gap is not the absence of DBSCAN, LSTM, or hybrid spatio-temporal models in the literature, but the limited availability of applied, reproducible, and interpretable workflows that connect hotspot detection, temporal forecasting, category-level analysis, and visual decision support at the district level.

Recent studies have increasingly examined not only model architecture but also the spatial unit used for crime forecasting. For example, hotspot-based and district-based partitioning approaches have been compared to evaluate how spatial representation affects forecasting performance. Multi-density hotspot forecasting has also been proposed to address heterogeneous crime concentrations across urban areas. These studies suggest that the choice of spatial unit and hotspot structure can substantially influence forecasting performance and operational interpretability.

Therefore, the present study positions Bekasi's district-level framework as an applied workflow that connects administrative units, DBSCAN-derived hotspot structures, and temporal forecasting outputs rather than as a new algorithmic architecture [36], [37].

Recent work on adaptive traffic volume and density estimation from legacy analog surveillance systems also highlights the importance of designing computational intelligence frameworks that remain interpretable, operationally useful, and applicable under constrained urban monitoring environments [38]. Although focused on traffic rather than crime forecasting, this line of work is relevant to the present study because both domains require spatio-temporal modelling, low-resource urban sensing, and decision-support outputs for public-safety planning.

D. CRIMINOLOGICAL THEORY AND MACHINE LEARNING ALIGNMENT

Integrating machine learning with criminological theory enhances the interpretability and applicability of such predictive models. Crime Pattern Theory by Brantingham & Brantingham [29], [30] posits that crime is not randomly distributed but occurs in predictable patterns influenced by human activity and environmental design, and supported by Weisburd [31]. Routine Activity Theory by Cohen & Felson [32] emphasizes the convergence of motivated offenders, suitable targets, and the absence of capable guardians as determinants of crime.

These theoretical frameworks align well with the designs of DBSCAN and LSTM. DBSCAN identifies repeatable spatial patterns, whereas LSTM captures behavioural routines and periodicities in temporal data. Previous studies have noted this synergy but have rarely demonstrated it using an operationally integrated forecasting tool. In this study, criminological theory is used as an interpretive lens rather than as a causal validation framework. The empirical findings are therefore discussed as being consistent with Crime Pattern Theory and Routine Activity Theory, not as definitive proof of these theories.

E. GAPS IN CURRENT RESEARCH

Although various studies support the use of spatial or temporal modelling, few have fully integrated both into a single predictive pipeline at the sub-city or district level. Most models do not address the limitations of imbalanced data, missing categories, or interpretability in operational policing contexts. However, the gap should be understood carefully: this study does not claim that hybrid spatio-temporal crime prediction is absent from prior literature. Rather, it addresses an applied gap in organizing established techniques into a district-level workflow for Bekasi City that combines hotspot identification, daily crime-count forecasting, exploratory category classification, and forecast-informed visualization.

Accordingly, the novelty of this study is applied and contextual rather than purely algorithmic. It lies in evaluating

how DBSCAN and LSTM, together with demographic enrichment and visual analytical outputs, can be structured into a decision-support workflow for local law-enforcement planning. This framing distinguishes the present work from studies that focus primarily on algorithmic improvement, city-level aggregate prediction, or isolated spatial or temporal modelling. The proposed workflow is therefore intended to provide context-specific evidence for district-level crime analysis, while explicitly acknowledging limitations related to data sparsity, class imbalance, generalizability, and ethical deployment.

III. METHODOLOGY

A. RESEARCH FRAMEWORK AND DESIGN

This study followed the Cross-Industry Standard Process for Data Mining (CRISP-DM) as illustrated in Figure 1, a well-established methodology for systematic data science projects. The CRISP-DM approach comprises six key stages: business understanding, data understanding, data preparation, modelling, evaluation, and deployment [33]. The framework ensures that technical efforts remain aligned with operational goals, in this case, supporting law enforcement agencies in proactive crime prevention. This study is situated within a quantitative, predictive modelling paradigm that leverages machine learning techniques to extract actionable insights from historical crime data. By adopting this structured process, the research maintains both methodological rigor and practical relevance.

B. DATA COLLECTION AND UNDERSTANDING

The original dataset contained 15,121 police records from Bekasi City. After sequential cleaning and filtering, including administrative boundary verification, removal of incomplete or inconsistent geocoded entries, duplicate checking, exclusion of non-street-crime cases, and standardization of district assignments, 11,555 valid street-crime incidents were retained as the final modelling dataset. This final dataset covered the period from January 19, 2020, to September 5, 2024, across 12 districts and 9 crime categories.

TABLE 1. Dataset processing summary.

Dataset stage	Number of records	Description
Original police records	15,121	Raw records obtained from Bekasi City Police
Excluded records	3,566	Records removed due to boundary mismatch, incomplete or inconsistent geocoding, duplicate entries, non-street-crime cases, or unresolved district assignment
Final cleaned modelling dataset	11,555	Valid geocoded street-crime incidents used for modelling and evaluation

For forecasting, the incident-level records were transformed into a district-day panel dataset. Each geocoded crime incident was first assigned to one of the 12 administrative districts. The number of incidents was then aggregated by district and date to obtain daily district-level crime counts. Missing district-date combinations were filled with zero counts to preserve a continuous daily time series for each district. This aggregation produced the temporal forecasting unit used by the LSTM and baseline models.

C. DATA CLEANING AND TRANSFORMATION

To ensure consistency and usability, several preprocessing tasks were applied: (1) Standardization of date formats across multiple years (resolving inconsistencies such as dd/mm/yy, mm-dd-yyyy, etc.). (2) Mapping multiple violated articles into standardized crime categories based on the Indonesian Criminal Code (KUHP): Removal of irrelevant and non-street crime types (e.g., cybercrime, domestic violence). (3) Conversion of location fields to numeric geospatial coordinates. (4) Resolution of missing data via logical imputation (e.g., interpolation of missing dates). (5) Transformation of field names to a consistent schema using snake_case conventions.

The final dataset includes essential fields such as crime_date, crime_latitude, crime_longitude, district, crime_category, and demographic enrichments such as district_population, and district_area_km2.

D. FEATURE ENGINEERING

To improve the model learning, a set of engineered features was derived.

Temporal Features: (1) Day of week, weekend flag, and month (to capture seasonal trends). (2) Lag features: crime count on previous day (t-1) and same weekday in previous week (t-7). (3) Rolling 7-day average crime count - Cyclical encodings using sine and cosine transformations for months and days to model periodicity. (4) Public holiday indicator (binary flag derived using Indonesian calendar).

Temporal Spatial Features: (1) Latitude and longitude. (2) DBSCAN-generated cluster labels for each incident (representing hotspot membership). (3) Distance to cluster centroid (used to assess centrality within a hotspot).

For district-level forecasting, incident-level spatial attributes were summarized into district-day representations. DBSCAN cluster labels were used to identify whether incidents belonged to historical hotspot structures, while distance-to-centroid values were used to characterize the proximity of incidents to hotspot centers. These spatial descriptors were aggregated at the district-day level and used together with temporal lag features and district-level demographic variables. This design allowed the forecasting model to retain spatial context while predicting daily crime counts at the administrative district level.

Demographic Features: (1) District-level population. (2) Total area per district (km²). (3) Population density (persons/km²).

These features enabled the model to learn both short-term patterns and long-term periodicities, while also factoring in the spatial concentration of incidents and the contextual influence of population distribution.

E. MODELING COMPONENTS

Three core modelling components were implemented: LSTM regression for district-level daily crime-count forecasting, DBSCAN for spatial hotspot identification, and crime-category classification using LSTM, Random Forest, and XGBoost classifiers.

LSTM Regression for Daily Crime Counts: The main forecasting task used LSTM neural networks to predict daily crime counts at the district level. Each model used a 14-day sliding window of historical data as input and predicted the next day's crime count. The models were trained separately for each district to capture localized patterns. The architecture included two LSTM layers (128 and 64 units), dropout regularization, and a dense output layer. The models were trained using the Adam optimizer and mean squared error (MSE) loss.

DBSCAN for Spatial Clustering: DBSCAN was used to identify high-density crime clusters across Bekasi City. The eps parameter was selected using k-distance analysis, where the elbow region of the k-distance curve was examined to identify a suitable neighborhood radius. The min_samples parameter was set to 10 to avoid forming clusters from very small and potentially unstable groups of incidents. To assess parameter robustness, sensitivity checking was conducted by comparing the number of clusters, the proportion of noise points, and the persistence of dominant hotspot structures across nearby eps and min_samples values. Clustering was performed on the full five-year dataset and on yearly subsets to distinguish persistent hotspots from transient or year-specific hotspot patterns.

Crime Category Classification: For the classification task, the original crime categories were consolidated into broader analytical groups to reduce sparsity and improve class interpretability. Extremely rare classes, such as categories with very few instances, were excluded only in one exploratory configuration to examine model behavior under severe class imbalance. This exclusion was not treated as the primary operational setting because it weakens direct comparability with the full-category classification problem. Therefore, the classification experiment was treated as a secondary exploratory analysis rather than the main contribution of the study.

F. MODEL EVALUATION STRATEGY AND LEAKAGE PREVENTION

To avoid temporal leakage, the forecasting dataset was split chronologically rather than randomly. The earliest 70% of observations were used for training, the subsequent 20% for validation, and the final 10% for testing. Feature scaling parameters were estimated only from the training set and then applied to the validation and test sets. Lag features

TABLE 2. LSTM forecasting configuration.

Component	Configuration
Forecasting target	Next-day district-level crime count
Spatial unit	District level
Temporal aggregation	Daily crime count
Input window	14 days
Forecast horizon	1 day ahead
Model type	Long Short-Term Memory (LSTM) neural network
Number of LSTM layers	2
Hidden units	128 and 64
Dropout regularization	0.20
Output layer	Dense layer with 1 output unit
Loss function	Mean Squared Error (MSE)
Optimizer	Adam
Learning rate	0.001
Batch size	32
Number of epochs	100
Early stopping	Patience = 10
Feature scaling	Min-Max normalization
Data split	Chronological 70% training, 20% validation, 10% testing
Leakage prevention	Scaling fitted only on training data; lag and rolling features computed from past observations only
Software environment	Python 3.11, TensorFlow/Keras, Scikit-learn, Pandas, NumPy
Computational environment	Google Colab with GPU support
Random seed	42

and rolling-window features were computed using only past observations available at each prediction time. This procedure ensured that future information was not used during model training or feature construction.

1) LSTM REGRESSION

The district-level forecasting task was evaluated using MAE, RMSE, and MAPE/sMAPE. MAE measures the average absolute difference between predicted and actual daily crime counts, while RMSE gives greater penalty to larger forecasting errors. MAPE and sMAPE were used to assess relative forecasting error. Because district-level daily crime counts are often sparse and may contain very small values, sMAPE was included as a complementary relative-error metric to reduce instability and avoid overinterpreting large percentage errors caused by near-zero actual values.

2) CRIME CATEGORY CLASSIFICATION

The crime-category classification task was evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Accuracy was reported to provide an overall measure of correctness, but it was not treated as the primary metric because the crime-category distribution was highly imbalanced. Precision, recall, and F1-score were used to assess class-level performance, especially for minority categories,

while confusion matrices were used to examine misclassification patterns across crime types.

G. TOOLS AND COMPUTATIONAL ENVIRONMENT

The experiments were implemented using Python 3.11. TensorFlow/Keras was used for LSTM modelling, Scikit-learn for preprocessing, classical machine-learning models, and evaluation metrics, and Pandas and NumPy for data manipulation. Matplotlib and Seaborn were used for statistical visualization, while Folium and HTML-based animation were used to generate geographic maps and forecast-informed spatial hotspot visualizations. Model development and training were conducted in Google Colab with GPU support.

H. REPRODUCIBILITY AND CODE AVAILABILITY

To support reproducibility, the modelling configuration, chronological data split, feature scaling procedure, leakage-prevention strategy, software environment, and random seed are reported in the methodology section. Hyperparameters for the LSTM model were selected based on validation-set performance and training stability, including the input window, number of LSTM units, dropout rate, learning rate, batch size, number of epochs, and early-stopping patience. DBSCAN parameters were selected using k-distance analysis and sensitivity checking across nearby eps and min_samples values by comparing the number of clusters, the proportion of noise points, and the persistence of dominant hotspot structures. Due to the sensitive nature of police-recorded crime data, the raw dataset cannot be publicly released. However, implementation details, pseudocode, and non-sensitive processing scripts may be provided upon reasonable request, subject to institutional and data-provider approval.

IV. RESULTS

A. PERFORMANCE OF LSTM-BASED FORECASTING BY DISTRICT

Forecasting performance was evaluated using MAE, RMSE, and sMAPE. The use of sMAPE is important because several districts have sparse daily crime counts, making percentage-based errors unstable when actual values are close to zero.

The LSTM models were applied independently to each of Bekasi's 12 districts to forecast daily crime counts. The models' effectiveness varied across districts, reflecting the differing stability of the crime trends in each region. Pondokmelati exhibited the best overall performance, with a Mean Absolute Error (MAE) of 0.24 and a Mean Absolute Percentage Error (MAPE) of 10.1%. This indicates that the predicted crime counts deviated, on average, by less than one incident per day from the actual observations, and the relative error remained small.

Districts such as Jatisampurna and Mustikajaya followed with favorable accuracy metrics (MAPE under 25%), showing that LSTM captured regular patterns effectively in areas

with consistent crime rates. In contrast, volatile districts, such as Bantargebang, demonstrated significantly higher MAPE ($>100\%$), largely because of their sporadic crime occurrences. On days with very low incident counts, even small absolute errors were translated into disproportionately high relative errors. This limitation is common in sparse datasets and illustrates the sensitivity of the percentage-based evaluation metrics.

Figure 2 presents a bar chart comparing the MAE, RMSE, and MAPE across the 12 districts. These metrics provide a multidimensional view of model performance, with MAE highlighting consistency, RMSE emphasizing the penalty for large deviations, and MAPE offering interpretability in proportional terms.

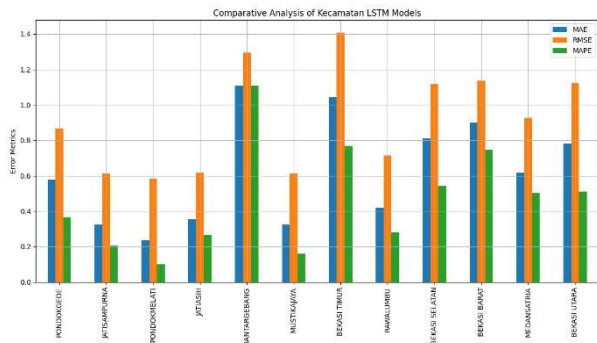


FIGURE 2. Comparative forecasting errors across districts.

Further analysis of the time-series plots for each district (Figure 3-5) revealed that the models closely tracked actual daily counts in predictable districts but struggled with outlier spikes.

In most cases, LSTM was able to replicate seasonal trends, such as crime increases during public holidays or weekends, which were captured via cyclical features. However, short-lived surges due to anomalous events (e.g., protests and emergencies) often eluded the model, suggesting the need for external variable integration in future iterations.

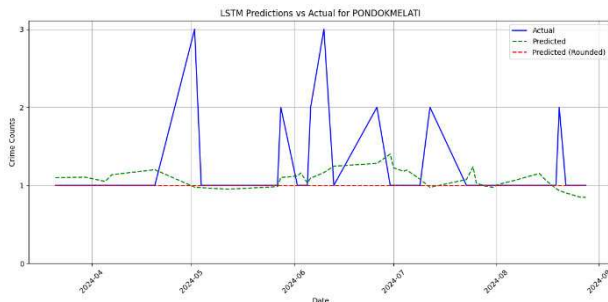


FIGURE 3. LSTM predictions vs actual visualization for pondokmelati.

For readability, only three representative district-level prediction plots are retained in the main manuscript: Pondokmelati as a relatively stable district, Pondokgede as a moderate-pattern district, and Bantargebang as a sparse

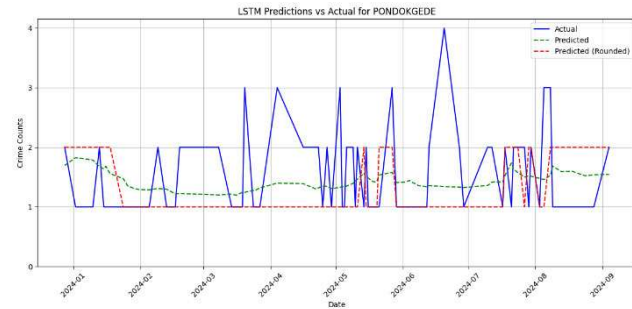


FIGURE 4. LSTM predictions vs actual visualization for pondokgede.

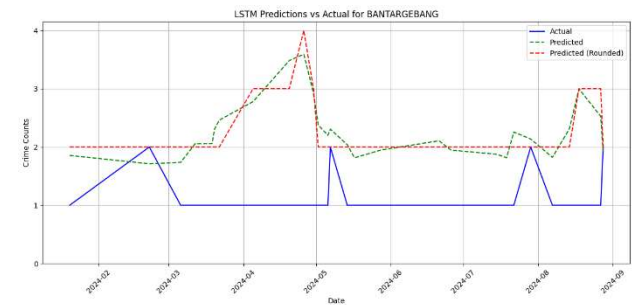


FIGURE 5. LSTM predictions vs actual visualization for bantargebang.

and volatile district. The remaining district-specific LSTM prediction-versus-actual plots are provided in the Supplementary Material.

B. BASELINE FORECASTING COMPARISON

Forecasting performance was evaluated using MAE, RMSE, and sMAPE. The use of sMAPE is important because several districts have sparse daily crime counts, making percentage-based errors unstable when actual values are close to zero.

TABLE 3. Baseline forecasting comparison for district-level daily crime counts.

Model	MAE	RMSE	sMAPE
Naïve persistence	0.769	1.22	88.768
7-day moving average	0.68	0.942	134.204
ARIMA/SARIMA	0.637	0.892	148.142
LSTM	0.636	0.892	147.817

The baseline comparison shows that the LSTM model achieved the lowest MAE, with only a very small margin over the ARIMA/SARIMA baseline, while its RMSE was comparable to that of ARIMA/SARIMA. However, LSTM did not achieve the lowest sMAPE; the naïve persistence baseline produced the lowest sMAPE value. This discrepancy reflects the instability of relative-error metrics under sparse district-level daily crime counts, where near-zero actual values can strongly affect percentage-based measures. Therefore, the LSTM results should be interpreted as

competitive in terms of absolute forecasting error rather than consistently superior across all evaluation metrics.

C. EVALUATION OF CRIME CATEGORY CLASSIFICATION MODELS

To forecast crime categories, six different experiments were conducted using LSTM, XGBoost, and Random Forest classifiers under various data balancing techniques. The classification task was particularly challenging because of the highly imbalanced distribution of crime types; for example, 'Fraud' accounted for more than 50% of the dataset, whereas categories such as 'Homicide' and 'Drug Offenses' were rare.

The baseline LSTM model trained on the imbalanced dataset defaulted to predicting the dominant class ('Fraud'), yielding a deceptively high accuracy of 51% but failing completely for minority classes. When balanced sampling was introduced, the model accuracy declined to 32%, but the macro F1-score improved, demonstrating a better distribution of predictive power across all classes. This discrepancy illustrates the inadequacy of accuracy as a standalone metric in highly imbalanced multi-class classification problems and highlights the importance of macro-averaged performance measures. In particular, improvements in macro F1-score provide a more reliable indication of whether minority classes are being meaningfully learned by the model.

The best-performing configuration was the XGBoost classifier with ADASYN oversampling and minority-class exclusion, achieving a macro-F1 score of 0.36 and an accuracy of 49.6%. However, this result should be interpreted cautiously. The exclusion of extremely rare minority classes was conducted only as an exploratory experiment to examine model behavior under severe class imbalance, and it weakens direct comparability with the full-category setting. Therefore, the crime-category classification component should be regarded as an exploratory secondary task rather than a mature operational classification module.

The relatively modest macro-F1 score of 0.36 indicates that category-level crime prediction remains challenging under the available feature representation and highly imbalanced class distribution. In particular, very rare categories such as homicide and drug offenses provide insufficient training examples for reliable supervised classification. Future work should investigate stronger imbalance-aware approaches, such as class-weighted learning, focal loss, balanced ensemble methods, hierarchical classification, and the incorporation of richer contextual variables.

TABLE 4. Classification performance summary across models.

Model	Accuracy (%)	Macro F1
LSTM (Imbalanced)	51.0	0.17
LSTM with Balanced Sampling	32.0	0.29
Hybrid LSTM + XGBoost	31.0	0.31
XGBoost + ADASYN (Minority Excluded)	49.6	0.36

The confusion matrices for each model illustrated the persistent misclassifications of less frequent categories into the majority class. Minority classes, such as 'Others,' were most frequently confused with 'Fraud', indicating insufficient distinguishing features. Despite feature engineering efforts (e.g., temporal lags and rolling means), further improvements will likely require the incorporation of contextual variables such as suspect profiles, event triggers, or location type meta-data. Accordingly, the classification results should not be used as an operational crime-category prediction module at this stage, but rather as preliminary evidence highlighting the difficulty of category-level forecasting under severe class imbalance.

D. DBSCAN-BASED SPATIAL HOTSPOT ANALYSIS

DBSCAN clustering revealed valuable insights into spatial crime patterns over a five-year period. A total of 30 clusters were identified using the optimized parameters ($\text{eps}=0.10$, $\text{min_samples}=10$). Cluster 0 alone accounted for over 10,000 incidents, representing long-standing high-risk zones in central and northern Bekasi. These persistent hotspots were consistent across all five years, affirming the Crime Pattern Theory's principle that certain geographic areas inherently attract crime due to target density and environmental layout.

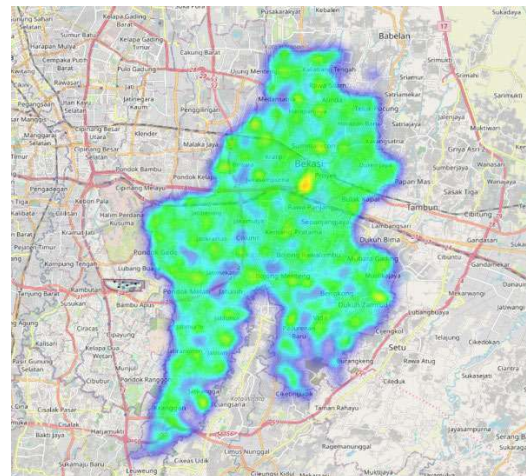


FIGURE 6. DBSCAN spatial clusters in bekasi (2020–2024).

Figure 6 shows the spatial distribution of the DBSCAN clusters overlaid on a map of Bekasi. Smaller clusters, ranging from 10 to 170 incidents, represented either seasonal hotspots (e.g., near event venues) or emerging crime zones. Their temporal volatility was further explored through year-by-year DBSCAN runs using variable eps thresholds (0.08 to 0.09), revealing transient clusters that surfaced only in specific years, such as during the COVID-19 lockdown or post-pandemic economic reopening.

The comparative analysis between yearly and cumulative clustering outcomes emphasized the value of multi-scale spatial analysis. Persistent clusters warrant long-term

interventions (e.g., fixed CCTV coverage), whereas transient clusters suggest the need for agile resource deployment.

E. FORECAST-INFORMED HOTSPOT VISUALIZATION

To support visual interpretation of forecasted crime risk, a 30-day LSTM crime-count forecast was combined with DBSCAN-derived historical hotspot proportions to produce forecast-informed spatial hotspot visualizations. This component is not intended as an independently learned spatial forecasting model. Instead, it serves as a decision-support visualization layer that allocates LSTM-predicted aggregate crime counts across historically observed hotspot structures to illustrate possible spatial risk patterns over time. The allocation process redistributed each day's predicted crime count across DBSCAN-derived clusters based on historical hotspot proportions. A limited random perturbation was used only for visualization purposes and was not treated as a learned spatial prediction mechanism. Hotspot intensities varied over the forecast window, illustrating recurring weekend peaks and midweek lulls that were consistent with the temporal patterns learned by the LSTM model. Therefore, the heatmap output should be interpreted as a forecast-informed visualization rather than a direct spatial forecast. Future work should replace this heuristic allocation procedure with a learned spatial forecasting model, such as graph-based or grid-based spatio-temporal modelling, to estimate spatial risk redistribution more rigorously.

Figure 7 shows three snapshots from the animation (Day 1, Day 15, and Day 30). High-risk areas remained consistent, whereas fluctuations in peripheral zones highlighted the need for adaptive patrol routing.



FIGURE 7. Forecast-informed crime risk visualization snapshots (day 1, Day 15, and day 30).

This visualization serves not only as a communication tool for police planners but also as a decision-support layer for daily patrol assignments. This allows supervisors to review possible geographic shifts in recorded crime risk and support earlier aggregate planning of personnel allocation.

F. CORRELATION ANALYSIS WITH DEMOGRAPHICS

To evaluate whether demographic pressure and economic vibrancy drive street-crime incidence, cumulative crime counts ($N = 11,555$) were aggregated for Bekasi's 12 districts from the 2020–2024 period. This aggregate was joined with district-level variables from Statistics Indonesia (Badan Pusat Statistik-BPS), specifically the 2023 mid-year population estimate and population density (persons/km²).

The study employed two correlation metrics: Pearson's r to measure linear association and Spearman's ρ as a rank-based robustness check against non-normality. Significance was assessed at $\alpha = 0.05$. Visual diagnostics include scatterplots with Ordinary Least Squares (OLS) trend lines (Figures 8 and 9) and a correlation heatmap (Figure 10).

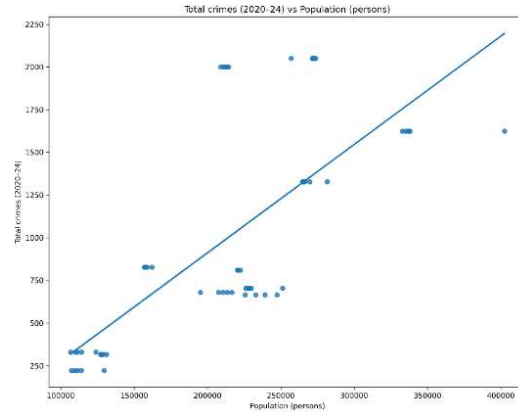


FIGURE 8. Total crimes vs population.

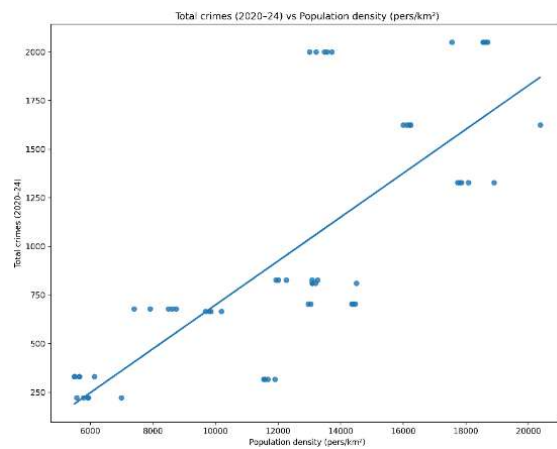


FIGURE 9. Total crimes vs population density.

As shown in Table 5, The correlation was stronger for population density (Pearson $r = 0.769$, $p = 0.003$; Spearman $\rho = 0.844$, $p = 0.001$). However, because the analysis was conducted across only 12 districts, these findings should be interpreted as contextual associations rather than causal or broadly generalizable evidence. The observed association is consistent with Routine Activity Theory, but it should not be interpreted as causal validation [32], suggesting that high population density facilitates a greater frequency of social interactions, thereby increasing the convergence of motivated offenders and suitable targets within the urban environment [24], [25], [29].

In summary, the results indicate that the integrated DBSCAN-LSTM workflow provides context-specific forecasting and visualization support for district-level crime analysis.

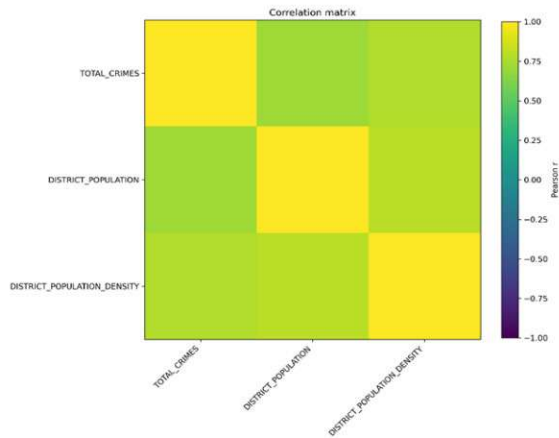


FIGURE 10. Correlation matrix heatmap.

TABLE 5. Summary of correlation results.

Variable	Pearson r	Pearson p -value	Spearman ρ	Spearman p -value
District Population	0.715	0.009	0.709	0.010
Population Density	0.769	0.003	0.844	0.001

G. ABLATION ANALYSIS OF FEATURE GROUPS

To examine the contribution of different feature groups, an ablation analysis was conducted by comparing several forecasting configurations: temporal features only, temporal features with DBSCAN-derived spatial features, temporal features with district-level demographic features, and the full configuration combining temporal, DBSCAN-derived spatial, and demographic features. This analysis was designed to assess whether spatial and demographic enrichment provides additional predictive value beyond temporal patterns alone.

The ablation results show that temporal features provide the primary predictive signal for district-level daily crime-count forecasting. Adding DBSCAN-derived spatial features and district-level demographic features produced only modest changes in MAE, RMSE, and sMAPE. The full feature configuration achieved the lowest sMAPE but did not uniformly outperform the reduced configurations in all metrics. These findings suggest that the added value of spatial and demographic features is context-dependent and should be interpreted as supporting operational interpretation rather than producing large uniform predictive gains.

V. CONCLUSION AND DISCUSSION

A. RESEARCH CONTRIBUTIONS AND PRACTICAL IMPLICATIONS

This study contributes an applied, data-driven framework for forecasting street-level crime using an integrated spatial-temporal modeling approach. By combining DBSCAN for hotspot detection and LSTM for daily temporal forecasting, the model provides context-specific analytical

support at the district level, which complements more generalized, macro-level crime prediction approaches.

The practical utility of the model lies in its ability to inform aggregate planning and decision-making in law enforcement. The persistent hotspots identified through DBSCAN reinforce the value of strategically placing permanent policing infrastructure, including surveillance cameras, patrol substations, and environmental design interventions aligned with Crime Prevention Through Environmental Design (CPTED) principles [29]. Meanwhile, transient and emerging hotspots offer a basis for agile deployment strategies such as short-term surge operations in response to rising risks.

The use of animated risk heatmaps provides an additional visual dimension that enhances operational briefings and interagency coordination. The daily LSTM forecasts may support aggregate patrol-planning discussions, especially in districts with stable temporal patterns. These analytical insights can support more informed resource-planning discussions while remaining subject to human oversight and ethical safeguards.

B. THEORETICAL AND ANALYTICAL IMPLICATIONS

The integration of the Crime Pattern Theory and Routine Activity Theory into the machine learning architecture strengthens the interpretability and theoretical rigor of the forecasting model. DBSCAN provides a spatial representation that is consistent with Crime Pattern Theory, whereas LSTM captures temporal regularities that can be interpreted in relation to Routine Activity Theory. The empirical results suggest that recorded crime patterns in Bekasi exhibit structured spatial and temporal regularities, but they should not be interpreted as causal validation of criminological theory.

This research also demonstrates how modern machine learning can serve not only as a predictive tool but also as a means of testing and extending criminological theory. The observed correlation between crime frequency and population density supports the idea that environmental exposure and demographic pressure are significant predictors of crime, warranting further exploration in future studies on urban criminology.

C. LIMITATIONS AND THREATS TO VALIDITY

This study has several limitations and threats to validity that should be considered when interpreting the findings. First, the analysis is based on police-recorded street-crime incidents from Bekasi City only. Therefore, the findings are context-specific and should not be generalized directly to other cities without retraining, local validation, and adjustment to different administrative, demographic, and policing conditions.

Second, the dataset covers the period from 2020 to 2024, which includes the COVID-19 pandemic and post-pandemic recovery period. These exceptional conditions may have affected mobility patterns, reporting behavior, economic activity, and crime occurrence. As a result, some temporal

TABLE 6. Ablation analysis of feature groups.

Feature setting	Temporal features	DBSCAN spatial features	Demographic features	MAE	RMSE	sMAPE
Temporal only	Yes	No	No	0.638	0.891	147.999
Temporal + DBSCAN	Yes	Yes	No	0.635	0.89	147.857
Temporal + demographic	Yes	No	Yes	0.635	0.89	148.143
Temporal + DBSCAN + demographic	Yes	Yes	Yes	0.636	0.892	147.817

patterns learned by the model may reflect period-specific disruptions rather than stable long-term crime dynamics.

Third, the dataset consists of reported and processed police records. Such data may be affected by reporting bias, enforcement intensity, underreporting, and differences in administrative recording practices across districts. Consequently, the model captures patterns in recorded crime rather than the complete underlying crime reality. Predictive-policing systems may amplify historical and societal biases when trained on police-recorded data that reflect uneven reporting, underreporting, or unequal enforcement intensity. Recent work on algorithmic fairness in predictive policing emphasizes the need for bias mitigation, fairness-aware evaluation, and human oversight before such systems are used in operational settings [39]. Therefore, the proposed workflow should be used only as an aggregate decision-support tool and should not be used for individual-level profiling, suspect prediction, or automated enforcement decisions.

Operational use of the proposed workflow requires careful governance. Forecast outputs should be reviewed by human analysts and should not be used as the sole basis for patrol deployment, resource allocation, or enforcement action. Because recorded crime data may reflect uneven reporting practices and enforcement intensity across districts, model outputs may reproduce existing institutional or societal biases if used without fairness-aware monitoring. Therefore, the workflow should be treated as an aggregate analytical aid for planning discussions, not as a predictive-policing system for automated or individual-level decision-making.

Fourth, district-level daily crime counts are often sparse and may contain very small values. This affects percentage-based metrics such as MAPE and can produce unstable relative errors when actual values are close to zero. For this reason, sMAPE was included as a complementary relative-error metric, and the forecasting results were interpreted with attention to district-level sparsity and volatility.

In addition, the forecasting outputs should be interpreted with uncertainty in mind. The current workflow reports point forecasts and error metrics, but it does not provide calibrated prediction intervals or probabilistic uncertainty estimates. Therefore, the forecasts should be used as approximate aggregate risk signals for planning discussions rather than as deterministic predictions of future crime occurrence.

Fifth, crime-category classification remains an exploratory component of this study. The best macro-F1 score remained modest, and one experimental configuration excluded extremely rare minority classes to examine model behavior under severe class imbalance. This limits direct comparability with the full-category classification setting and indicates that category-level prediction should not yet be treated as an operationally mature module.

Sixth, the forecast-informed hotspot visualization should not be interpreted as an independently learned spatial forecasting model. The visualization allocates LSTM-predicted aggregate crime counts across DBSCAN-derived historical hotspot proportions. Therefore, it should be understood as a decision-support visualization layer rather than as a direct spatial prediction mechanism.

Finally, the demographic correlation analysis was conducted across only 12 districts. Although the observed correlations suggest contextual associations between population density and recorded crime incidence, they do not establish causality or provide generalizable validation of criminological theory. Future studies should incorporate broader spatial samples, richer socioeconomic variables, and cross-city validation to strengthen external validity. The workflow may be transferable as a general analytical structure, but its trained parameters, hotspot patterns, and forecasting performance should not be assumed to generalize beyond Bekasi City without local data, retraining, and validation.

D. RECOMMENDATIONS FOR FUTURE RESEARCH

Future research should extend the proposed workflow in several directions. First, real-time forecasting can be explored by integrating live data streams from law-enforcement information systems, enabling more dynamic model updating and faster analytical feedback. Second, the incorporation of exogenous variables, such as socioeconomic indicators, mobility patterns, weather conditions, public events, and land-use characteristics, may help explain irregular crime spikes and improve the robustness of anomaly detection.

Further methodological development is also needed. Graph Neural Networks, transformer-based temporal models, and attention mechanisms may provide stronger representations of spatial dependencies and long-range temporal patterns. In addition, coupling the forecasting workflow with agent-based simulation may help evaluate the

potential effects of different interventions, such as additional patrol allocation, environmental design changes, or preventive community programs, before implementation in real-world settings.

Future studies should also examine cross-city transferability. Testing the framework in cities with different demographic structures, crime profiles, urban forms, and policing practices would help assess its scalability and external validity. Finally, responsible deployment should remain a central direction for future work. Explainable AI techniques such as SHAP and LIME may improve transparency and interpretability [34], [35], while public documentation, audit trails, fairness monitoring, and clear accountability mechanisms are needed to ensure that predictive analytics supports public safety without reinforcing bias or misuse [28]. Future operational deployment should also consider lightweight, calibrated, and secure AI design principles, particularly when predictive models are used in resource-constrained and high-accountability environments [40].

E. CONCLUSION

This study shows that combining spatial clustering and temporal deep learning can provide a useful decision-support workflow for district-level crime forecasting. The DBSCAN-LSTM framework offers operational advantages through granular insights, interpretable risk maps, and a theory-informed interpretation. The workflow can support aggregate planning, visual interpretation, and more informed discussion of urban public-safety patterns when used with appropriate human oversight and ethical safeguards.

Future advancements should prioritize real-time adaptability, model transparency, and ethical safeguards to ensure that technological innovation in policing contributes constructively to community safety and social equity.

REFERENCES

- [1] R. A. Berk and J. Bleich, "Statistical procedures for forecasting criminal behavior: A comparative assessment," *Criminology Public Policy*, vol. 12, no. 3, pp. 513–544, Aug. 2013.
- [2] A. F. Mohammed and W. R. Baiee, "The GIS based criminal hotspot analysis using DBSCAN technique," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 928, no. 3, Nov. 2020, Art. no. 032081, doi: [10.1088/1757-899x/928/3/032081](https://doi.org/10.1088/1757-899x/928/3/032081).
- [3] F. Dakalbab, M. Abu Talib, O. Abu Waraga, A. Bou Nassif, S. Abbas, and Q. Nasir, "Artificial intelligence & crime prediction: A systematic literature review," *Social Sci. Humanities Open*, vol. 6, no. 1, 2022, Art. no. 100342, doi: [10.1016/j.ssaho.2022.100342](https://doi.org/10.1016/j.ssaho.2022.100342).
- [4] M. M. Mukto, M. Hasan, M. M. Al Mahmud, I. Haque, M. A. Ahmed, T. Jabid, M. S. Ali, M. R. A. Rashid, M. Manzurul Islam, and M. Islam, "Design of a real-time crime monitoring system using deep learning techniques," *Intell. Syst. Appl.*, vol. 21, Mar. 2024, Art. no. 200311.
- [5] W. L. Perry, B. McInnis, C. C. Price, S. C. Smith, and J. S. Hollywood, *Predictive Policing: The Role of Crime Forecasting in Law Enforcement Operations*. Santa Monica, CA, USA: RAND Corporation, 2013.
- [6] E. Groff and N. La Vigne, "Forecasting the future of predictive crime mapping," *Crime Prevention Stud.*, vol. 13, pp. 29–58, 2002.
- [7] M. B. Chamlin, "Crime and arrests: An autoregressive integrated moving average (ARIMA) approach," *J. Quant. Criminology*, vol. 4, no. 3, pp. 247–258, Sep. 1988, doi: [10.1007/bf01072452](https://doi.org/10.1007/bf01072452).
- [8] P. Chen, H. Yuan, and X. Shu, "Forecasting crime using the ARIMA model," in *Proc. 5th Int. Conf. Fuzzy Syst. Knowl. Discovery*, Oct. 2008, pp. 627–630, doi: [10.1109/fskd.2008.222](https://doi.org/10.1109/fskd.2008.222).
- [9] K. Islam and A. Raza, "Forecasting crime using ARIMA model," 2020, *arXiv:2003.08006*.
- [10] C. V. Redoblo, J. L. G. Redoblo, R. A. Salmingo, C. M. Padilla, and J. C. T. Arroyo, "Forecasting the influx of crime cases using seasonal autoregressive integrated moving average model," *Int. J. Adv. Appl. Sci.*, vol. 10, no. 8, pp. 158–165, Aug. 2023, doi: [10.21833/ijaas.2023.08.018](https://doi.org/10.21833/ijaas.2023.08.018).
- [11] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997, doi: [10.1162/neco.1997.9.8.1735](https://doi.org/10.1162/neco.1997.9.8.1735).
- [12] S. Hochreiter, "The vanishing gradient problem during learning recurrent neural nets and problem solutions," *Int. J. Uncertainty, Fuzziness Knowl.-Based Syst.*, vol. 6, no. 2, pp. 107–116, Apr. 1998, doi: [10.1142/s0218488598000094](https://doi.org/10.1142/s0218488598000094).
- [13] A. Graves and J. Schmidhuber, "Framewise phoneme classification with bidirectional LSTM and other neural network architectures," *Neural Netw.*, vol. 18, nos. 5–6, pp. 602–610, Jul. 2005, doi: [10.1016/j.neunet.2005.06.042](https://doi.org/10.1016/j.neunet.2005.06.042).
- [14] C. Ubal, G. Di-Giorgi, J. E. Contreras-Reyes, and R. Salas, "Predicting the long-term dependencies in time series using recurrent artificial neural networks," *Mach. Learn. Knowl. Extraction*, vol. 5, no. 4, pp. 1340–1358, Oct. 2023, doi: [10.3390/make5040068](https://doi.org/10.3390/make5040068).
- [15] X. Wang, M. S. Gerber, and D. E. Brown, "Spatio-temporal modeling of criminal incidents using LSTM networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 10, pp. 3870–3880, Oct. 2019.
- [16] H.-W. Kang and H.-B. Kang, "Prediction of crime occurrence from multi-modal data using deep learning," *PLoS ONE*, vol. 12, no. 4, Apr. 2017, Art. no. e0176244.
- [17] G. Hajela, M. Chawla, and A. Rasool, "A clustering based hotspot identification approach for crime prediction," *Proc. Comput. Sci.*, vol. 167, pp. 1462–1470, 2020.
- [18] M. Ester, H. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. 2nd Int. Conf. Knowl. Discovery Data Mining (KDD)*, 1996, pp. 226–231.
- [19] C. Yu, M. D. Ward, M. Morabito, and W. Ding, "Crime forecasting using spatio-temporal pattern with DBSCAN," in *Proc. IEEE Int. Conf. Data Sci. Adv. Anal. (DSAA)*, 2016, pp. 570–579.
- [20] M. Leitner and M. Helbich, "The impact of hurricanes on crime: A spatio-temporal analysis in the city of Houston, Texas," *Cartography Geographic Inf. Sci.*, vol. 38, no. 2, pp. 213–221, Jan. 2011.
- [21] S. Chainey and J. Ratcliffe, *GIS and Crime Mapping*. Hoboken, NJ, USA: Wiley, 2005.
- [22] J. Zhao, X. Tang, and X. Zhou, "Crime prediction model based on spatial clustering and ARIMA model," *J. Quant. Criminology*, vol. 36, no. 2, pp. 415–436, 2020.
- [23] M. Shan, C. Ye, P. Chen, and S. Peng, "Ada-GCNLSTM: An adaptive urban crime spatiotemporal prediction model," *J. Saf. Sci. Resilience*, vol. 6, no. 2, pp. 226–236, Jun. 2025.
- [24] Y. Du and N. Ding, "A systematic review of multi-scale spatio-temporal crime prediction methods," *ISPRS Int. J. Geo-Inf.*, vol. 12, no. 6, p. 209, May 2023, doi: [10.3390/ijgi12060209](https://doi.org/10.3390/ijgi12060209).
- [25] O. Kounadi, A. Ristea, A. Araujo, and M. Leitner, "A systematic review on spatial crime forecasting," *Crime Sci.*, vol. 9, no. 1, p. 7, May 2020, doi: [10.1186/s40163-020-00116-7](https://doi.org/10.1186/s40163-020-00116-7).
- [26] P. Stalidis, T. Semertzidis, and P. Daras, "Examining deep learning architectures for crime classification and prediction," *Forecasting*, vol. 3, no. 4, pp. 741–762, Oct. 2021, doi: [10.3390/forecast3040046](https://doi.org/10.3390/forecast3040046).
- [27] J. Cohen, W. L. Gorr, and A. M. Olligschlaeger, "Leading indicators and spatial interactions: A crime-forecasting model for proactive police deployment," *Geographical Anal.*, vol. 39, no. 1, pp. 105–127, Jan. 2007, doi: [10.1111/j.1538-4632.2006.00697.x](https://doi.org/10.1111/j.1538-4632.2006.00697.x).
- [28] A. Malik, R. Maciejewski, S. Towers, S. McCullough, and D. S. Ebert, "Proactive spatiotemporal resource allocation and predictive visual analytics for community policing and law enforcement," *IEEE Trans. Vis. Comput. Graph.*, vol. 20, no. 12, pp. 1863–1872, Dec. 2014, doi: [10.1109/TVCG.2014.2346926](https://doi.org/10.1109/TVCG.2014.2346926).
- [29] P. J. Brantingham and P. L. Brantingham, *Environmental Criminology*. Newbury Park, CA, USA: Sage, 1981.
- [30] P. J. Brantingham and P. L. Brantingham, *Patterns in Crime*. New York, NY, USA: Macmillan, 1984.
- [31] D. Weisburd, "The law of crime concentration and the criminology of place," *Criminology*, vol. 53, no. 2, pp. 133–157, 2015.

- [32] L. E. Cohen and M. Felson, "Social change and crime rate trends: A routine activity approach," *Amer. Sociol. Rev.*, vol. 44, no. 4, pp. 588–608, Aug. 1979.
- [33] R. Wirth and J. Hipp, "CRISP-DM: Towards a standard process model for data mining," in *Proc. 4th Int. Conf. Practical Appl. Knowl. Discovery Data Mining*, 2000, pp. 29–40.
- [34] F. Ersöz, T. Ersöz, F. Marcelloni, and F. Ruffini, "Artificial intelligence in crime prediction: A survey with a focus on explainability," *IEEE Access*, vol. 13, pp. 59646–59674, 2025, doi: [10.1109/ACCESS.2025.3553934](https://doi.org/10.1109/ACCESS.2025.3553934).
- [35] P. Hermosilla, S. Berrios, and H. Allende-Cid, "Explainable AI for forensic analysis: A comparative study of SHAP and LIME in intrusion detection models," *Appl. Sci.*, vol. 15, no. 13, p. 7329, Jun. 2025, doi: [10.3390/app15137329](https://doi.org/10.3390/app15137329).
- [36] E. Cesario, P. Lindia, and A. Vinci, "Comparing machine learning-based crime hotspots versus police districts: What's the best approach for crime forecasting?" *IEEE Access*, vol. 13, pp. 133053–133077, 2025, doi: [10.1109/ACCESS.2025.3592668](https://doi.org/10.1109/ACCESS.2025.3592668).
- [37] E. Cesario, P. Lindia, and A. Vinci, "Multi-density crime predictor: An approach to forecast criminal activities in multi-density crime hotspots," *J. Big Data*, vol. 11, no. 1, May 2024, doi: [10.1186/s40537-024-00935-4](https://doi.org/10.1186/s40537-024-00935-4).
- [38] D. O. D. Handayani, T. Mantoro, F. Arifman, B. Rahman, M. A. Ayu, and R. T. Handayanto, "A canonical computational intelligence framework for adaptive traffic volume and density estimation from legacy analog surveillance systems," *IEEE Access*, vol. 14, pp. 55707–55721, 2026, doi: [10.1109/ACCESS.2026.3682231](https://doi.org/10.1109/ACCESS.2026.3682231).
- [39] A. S. Almasoud and J. A. Idowu, "Algorithmic fairness in predictive policing," *AI Ethics*, vol. 5, no. 3, pp. 2323–2337, Jun. 2025, doi: [10.1007/s43681-024-00541-3](https://doi.org/10.1007/s43681-024-00541-3).
- [40] S. Karthika, J. K. Gnanaselvi, S. N. Sellappan, M. Gokilavani, and H. Koten, "TrustEdgeAI: A lightweight, calibrated, and secure deep learning framework for intrusion detection in edge-IoT environments," *Indonesian J. Mach. Learn. Intell. Syst.*, vol. 1, no. 1, pp. 14–21, 2026.



NANAN SOEKARNA received the Master's of Computer Science degree from Nusa Putra University, Indonesia, and the degree from the Indonesian Police Academy, in 1978. He is currently a retired Senior Officer of Indonesian National Police (Polri) and the Chairman of the Board of Trustees of Universitas Pendidikan Indonesia, from 2025 to 2030. He is also the Vice Chair of the Forum of Boards of Trustees for State Universities with Legal Entity Status (PTNBH), from 2025 to 2026. During his police career, he held strategic leadership roles, including the Deputy Chief of the Indonesian National Police, the Head of the Public Relations Division, and the Inspector General of Supervision. He received the Adhi Makayasa Award.



TEDDY MANTORO (Senior Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in computer science from the School of Computer Science, The Australian National University (ANU), Canberra, Australia. He is currently a Computer Science Professor with Nusa Putra University, Sukabumi, Indonesia. He has conducted intensive work in an intelligent environment that uses computational intelligence. He developed the concept and theory of context-aware computing

for the intelligent environment, and as a proof of concept, he and his laboratory developed many prototypes that led to many awards. His research interests include information security, pervasive computing, intelligent environment/the IoT, artificial intelligence, deep learning, information security, wireless sensor networks, and context-aware-computing. He received five Gold, nine Silver, and 11 Bronze medals from the National and International IT Innovation Competitions, since 2009.



UMAR ADITIAWARMAN received the Ph.D. degree in information and communication technology from International Islamic University Malaysia (IIUM), Kuala Lumpur, Malaysia. He is currently the Deputy Director and the Head of the Halal Value Chain Division, Komite Nasional Ekonomi dan Keuangan Syariah (KNEKS), Indonesia, where he focuses on digital transformation for the Islamic economy and halal industry. He is also a Lecturer with Nusa Putra University, Indonesia. His professional experience includes directorship at the Centre for IT Advancement–IIUM and managerial roles in ICT services and international business development. His research interests include information security management, digital governance, blockchain applications, and applied ICT for public-sector and industry systems.



FIRMAN ARIFMAN (Member, IEEE) received the Associate Diploma degree in information systems from TAFE South Australia, the Bachelor of Business degree in management information systems from the University of South Australia, Adelaide, Australia, the Graduate Certificate degree in project management from RMIT University, Melbourne, Australia, and the Master's of Computer Science degree in artificial intelligence from Nusa Putra University (NPU), Indonesia, where he is currently pursuing the Master's of Computer Science degree in cybersecurity and the Ph.D. degree in computer science. His research and professional interests include applied artificial intelligence, cybersecurity analytics, scalable system architecture, and the integration of data-driven models into operational and policy-relevant environments.



BEN RAHMAN received the bachelor's degree in information systems, the master's degree in information systems management, and the Ph.D. degree in computer science from Universitas Bina Nusantara, Indonesia, in 2012, 2017, and 2024, respectively. Since 2018, he has been a Lecturer with Universitas Nasional, Jakarta. Prior to his academic career, he worked in international and multinational industries across multiple countries, contributing practical engineering and enterprise system experience to his research and teaching. He is a researcher in computational intelligence specializing in adaptive AI systems and industrial machine learning. He has developed intelligent web, mobile, and cloud-based systems and has published in nationally and internationally indexed journals. He also serves as a reviewer for several national and international scientific journals. His research interests include computational intelligence, adaptive AI systems, generative AI, and machine learning for scalable industrial applications.

...